DeepTune

Efficient Layer-wise Feature Extraction and Fine-Tuning of Foundation Models

Abdelrhman Werby, Jil Panter, Sejal Jadhav
AndAnotherTeam

Modality 2

- Week 1
- Week 2
- Week 3
- Week 4
- Week 5
- Week 6
- Week 7
- Week 8
- Week 9
- Week 10
- Bonus
- Literature

Motivation

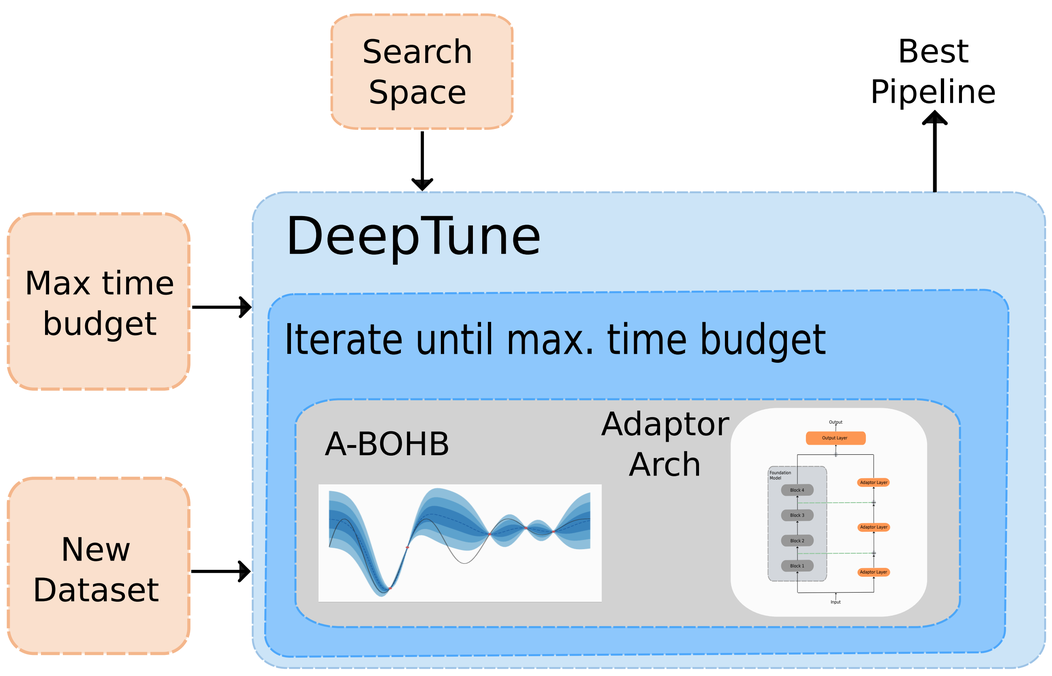
- Pretrained foundation models have revolutionized CV.
- Transfer learning from these models yields strong CV performance.
- Fine-tuning with adaptors is gaining attention for its efficiency.
- How to select optimal layers for adaptor addition and feature extraction?

Contributions

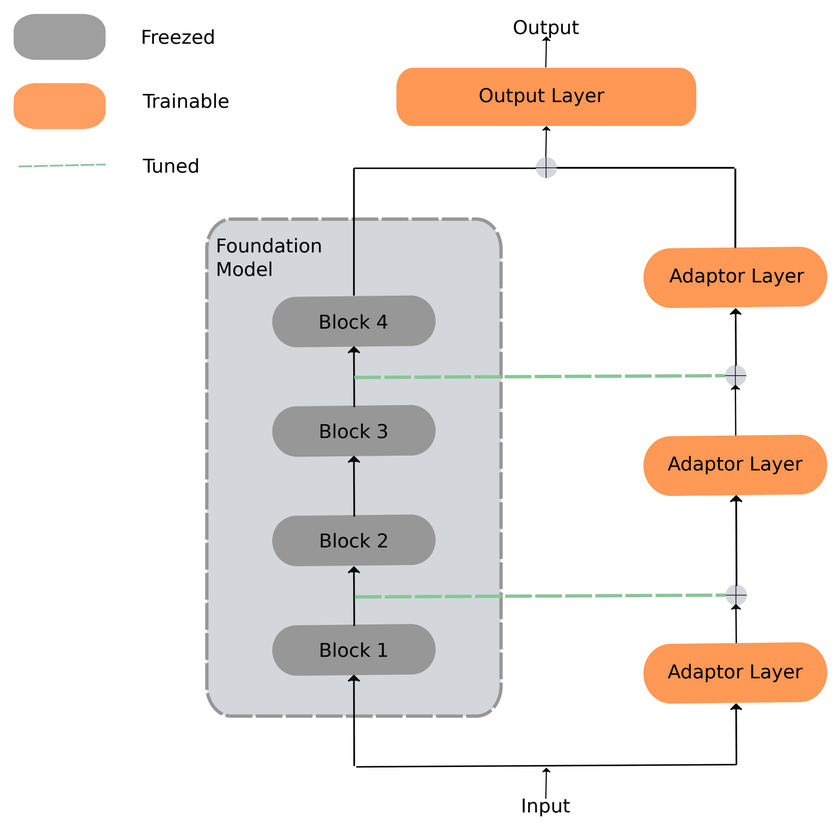
- We present effective pipeline for selecting the optimal layer to add adaptor layers.
- We design a search space covering common fine-tuning strategies.
- We compare our approach against multiple baselines, including common fine-tuning strategies and state-of-the-art HPO methods.

Our Approach

DeepTune



Efficient Adaptors



Resources Used

For development:
- 1 RTX3090 GPU
- 1 Ryzen9 5900x
- Total Compute Time 1000 CPU-h

For AutoML:
- 1 RTX3090 GPU
- 48 GPU-h

Workforce:
- 2 full-weeks

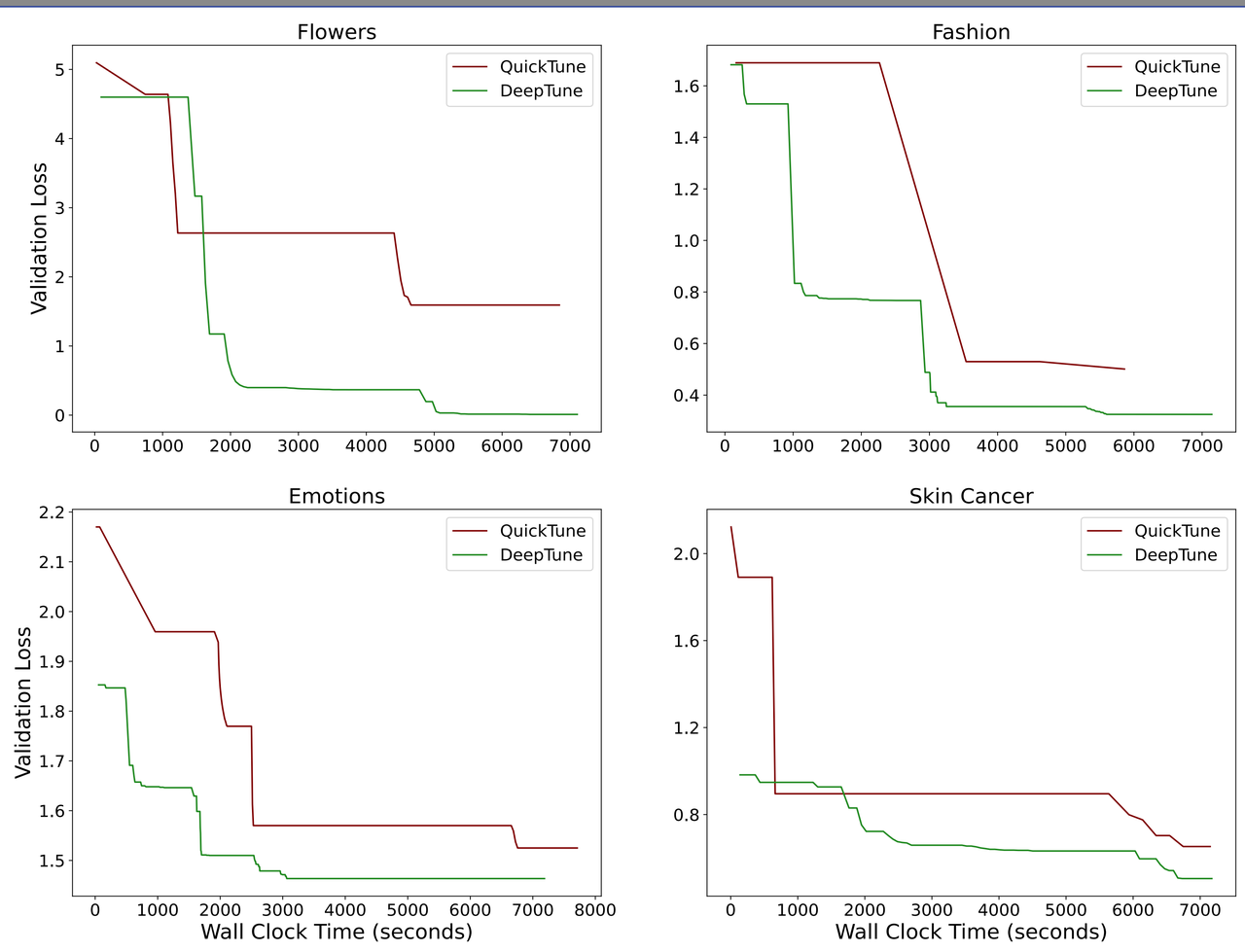
Empirical Results

Table 1: Comparison against efficient finetuning

Model	Fashion	Emotions	Flowers	Skin Cancer
QuickTune	0.78	0.32	0.14	0.75
Dinov2 + LP	0.83	0.37	**1.0**	0.60
DeepTune	**0.88**	**0.44**	0.99	**0.80**

Table 2: Search Space Summary

Hyperparameter Group	Hyperparameter
Fine-Tuning Strategies	Intermediate Layers*
Optimizer	Type (Adam, Momentum, SGD, AdamW), Betas*, Learning Rate*, Weight Decay*, Batch Size*, Momentum*
Scheduler	Type (None, Step, Multistep, Plateau, Cosine), Step Size*, Milestones*, Patience Epochs*, Factor*, Mode*, T Max*, eta Min*



Number of queries for test score generation: **1**

References

[1] Arango, Sebastian Pineda, et al. Quick-Tune: Quickly Learning Which Pretrained Model to Finetune and How. arXiv:2306.03828, arXiv, 22 Feb. 2024. arXiv.org, http://arxiv.org/abs/2306.03828.

[2] Houlsby, Neil, et al. Parameter-Efficient Transfer Learning for NLP. arXiv:1902.00751, arXiv, 13 June 2019. arXiv.org, http://arxiv.org/abs/1902.00751.

[3] Klein, Aaron, et al. Model-Based Asynchronous Hyperparameter and Neural Architecture Search. arXiv:2003.10865, arXiv, 30 June 2020. arXiv.org, http://arxiv.org/abs/2003.10865.

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